

CSCE 190: Computing in the Modern World

Class 3 – Sharing Resources via GitHub

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

AUG/ SEP 2026

Carolinian Creed: “I will practice personal and academic integrity.”

Credits: Copyrights of all material reused acknowledged

Outline

- Introduction Section
 - Recap from Last Class
- Main Section
 - Sharing Resources – Importance and What to Watch Out For
 - GitHub
 - GitHub pages
- Conclusion Section
 - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 2

We talked about

- Understanding an Organization
- Understanding Jobs and Career Success
 - Research Career
 - Money v/s Success
- From Course to Careers
- Course Advising

Activity Given – Writing Mission Statement

- Objective: identifying your **personal** mission for a career and success
- Guideline:
 - Think of short (1-3 years), medium (4-7 years) and long-term (8 -) goals
 - While career is the focus, feel free to include personal goals as well
 - Define success criteria for all goals
 - Try to structure how short term (e.g., undergrad program) may help meet **medium-term goals**
 - Try to structure how short and medium may help meet **long-term goals**
- Try to identify organizations and job roles of interest
- Strategize how your major and courses may help

Main Section

Sharing of Resources - Importance

- Growing the ecosystem
 - Collaboration: doing more and faster together
 - Learning: see and emulate from others
- Raising one's own professional stature
 - More the users, more the value of resource (asset) provided

Sharing of Resources – What to Watch Out For

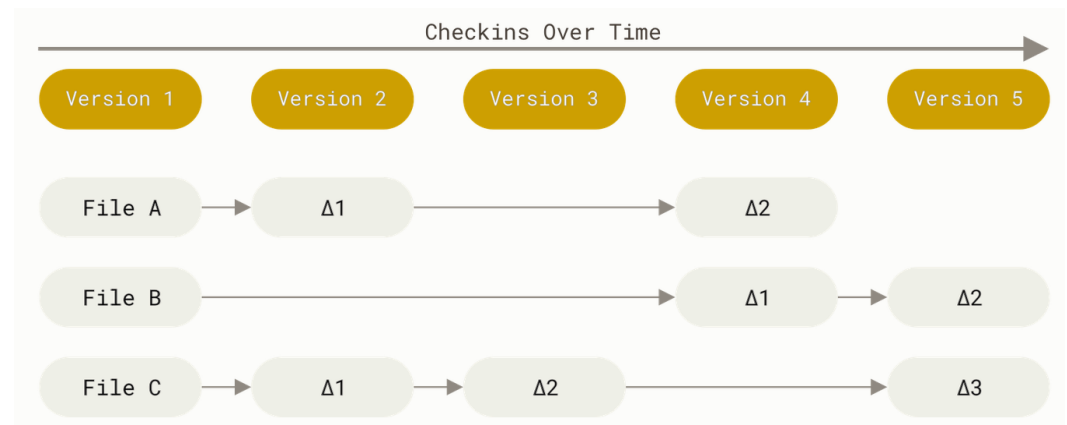
- Intellectual property (IP)
- Privacy
- Unintended commitments

History – Git, GitHub and Significance

- Git
 - Perform version control for files using a (remote, local) server
 - Created by Linus Torvalds (creator of Linux), 2005 (Source: <https://en.wikipedia.org/wiki/Git>)
- GitHub
 - Running Git, remote server maintained by a organization (GitHub). Created in 2008.
 - Originally open-source and non-profit, sold in 2018 to Microsoft
- Significance
 - GitHub is the world's largest [source code](https://en.wikipedia.org/wiki/GitHub) host as of June 2023. Over five billion developer contributions were made to more than 500 million open source projects in 2024.(Source: <https://en.wikipedia.org/wiki/GitHub>)
 - Works with **any type of files**
- We will use Git and GitHub terms interchangeably for convenience

Git Concepts

- File – anything worth storing
 - Repository – a place with files and folders
- Versions – a file state at a given time
- Commit - taking a snapshot of a repo; push, pull, status
- Branch

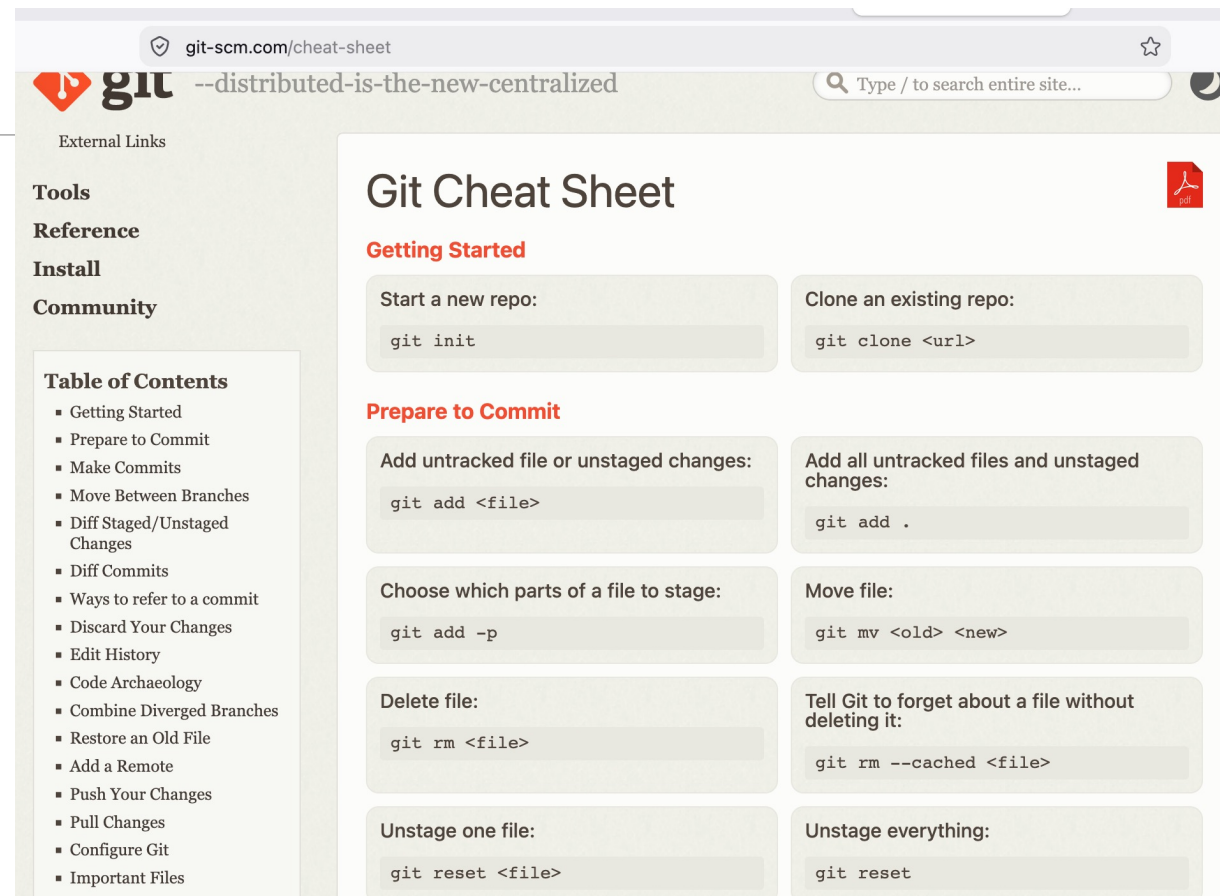


Free book for further reading:

- **Pro Git**, Scott Chacon and Ben Straub (2014), <https://git-scm.com/book/en/v2>
- Note license: [Creative Commons Attribution Non Commercial Share Alike 3.0 license](https://creativecommons.org/licenses/by-nc-sa/3.0/)

Git Resources

- Cheat sheet:
<https://git-scm.com/cheat-sheet>
- Videos:
<https://git-scm.com/videos>



The screenshot shows the Git Cheat Sheet website. The browser address bar displays 'git-scm.com/cheat-sheet'. The website header includes the Git logo, the slogan '--distributed-is-the-new-centralized', and a search bar. A left sidebar contains navigation links: 'External Links', 'Tools', 'Reference', 'Install', 'Community', and a 'Table of Contents' with a list of topics. The main content area is titled 'Git Cheat Sheet' and is organized into sections: 'Getting Started' (with 'Start a new repo:' and 'Clone an existing repo:'), 'Prepare to Commit' (with 'Add untracked file or unstaged changes:', 'Add all untracked files and unstaged changes:', 'Choose which parts of a file to stage:', 'Move file:', 'Delete file:', 'Tell Git to forget about a file without deleting it:', 'Unstage one file:', and 'Unstage everything:'). Each section provides a brief description and the corresponding Git command in a text box.

git-scm.com/cheat-sheet

--distributed-is-the-new-centralized

Type / to search entire site...

External Links

Tools

Reference

Install

Community

Table of Contents

- Getting Started
- Prepare to Commit
- Make Commits
- Move Between Branches
- Diff Staged/Unstaged Changes
- Diff Commits
- Ways to refer to a commit
- Discard Your Changes
- Edit History
- Code Archaeology
- Combine Diverged Branches
- Restore an Old File
- Add a Remote
- Push Your Changes
- Pull Changes
- Configure Git
- Important Files

Git Cheat Sheet

Getting Started

Start a new repo:

```
git init
```

Clone an existing repo:

```
git clone <url>
```

Prepare to Commit

Add untracked file or unstaged changes:

```
git add <file>
```

Add all untracked files and unstaged changes:

```
git add .
```

Choose which parts of a file to stage:

```
git add -p
```

Move file:

```
git mv <old> <new>
```

Delete file:

```
git rm <file>
```

Tell Git to forget about a file without deleting it:

```
git rm --cached <file>
```

Unstage one file:

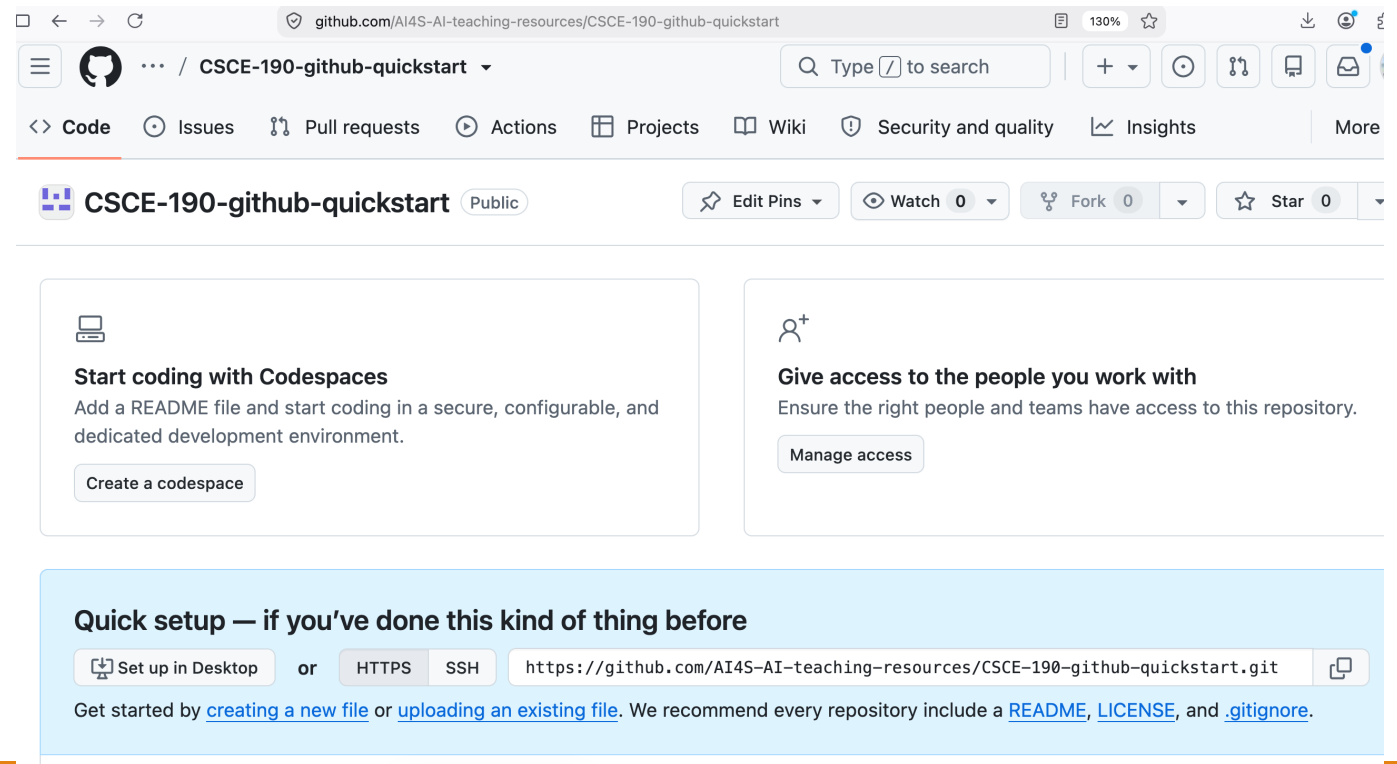
```
git reset <file>
```

Unstage everything:

```
git reset
```

GitHub Concepts

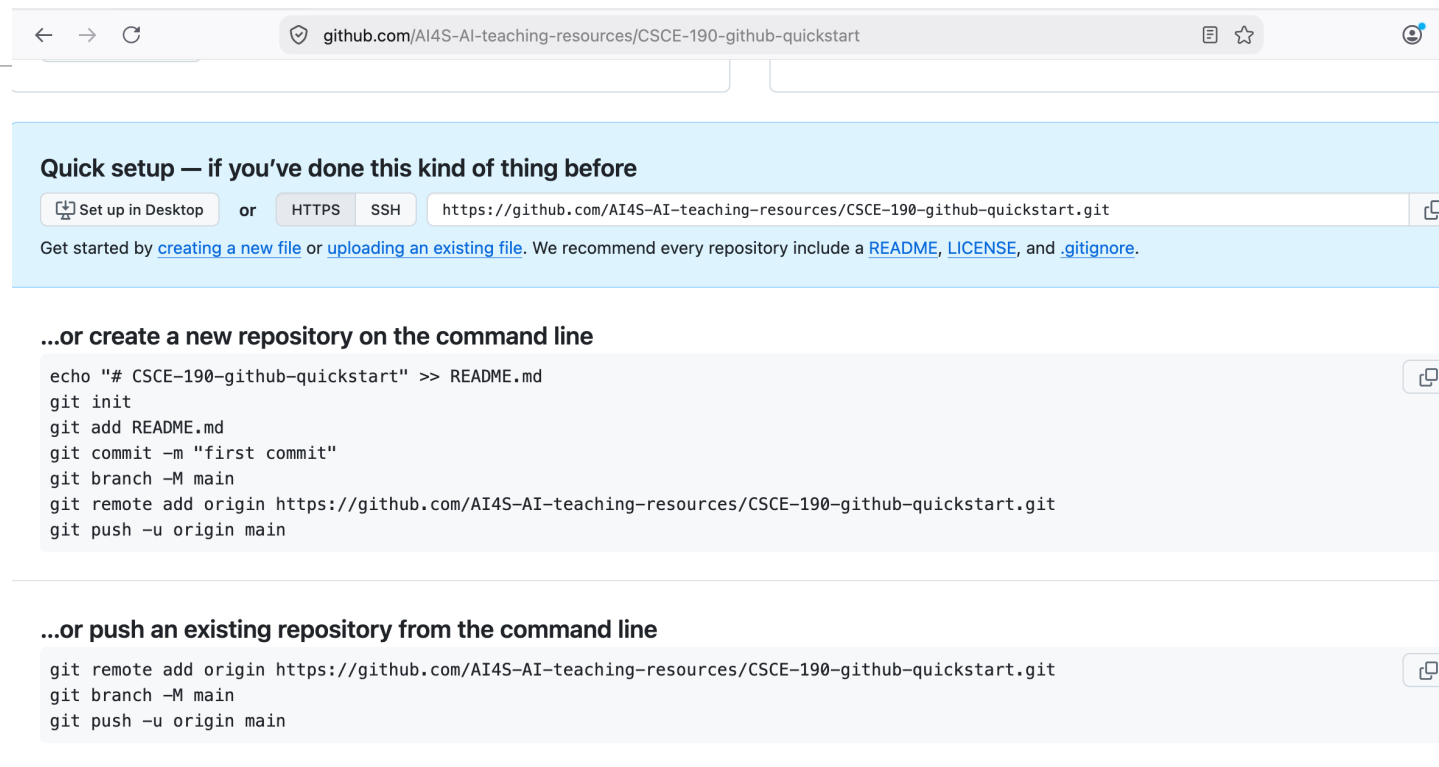
- Organizations
- Projects
- Team



GitHub Concepts

Ways to operate

- Command-line
- GitHub tool
- Using browser



The screenshot shows the GitHub repository page for `github.com/AI4S-AI-teaching-resources/CSCE-190-github-quickstart`. The page is titled "Quick setup — if you've done this kind of thing before". It provides two main options for setting up the repository: "Set up in Desktop" and "HTTPS" (selected). The HTTPS URL is `https://github.com/AI4S-AI-teaching-resources/CSCE-190-github-quickstart.git`. Below this, it says "Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#)."

Below the quick setup section, there are two sections for creating or pushing a repository from the command line.

...or create a new repository on the command line

```
echo "# CSCE-190-github-quickstart" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/AI4S-AI-teaching-resources/CSCE-190-github-quickstart.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/AI4S-AI-teaching-resources/CSCE-190-github-quickstart.git
git branch -M main
git push -u origin main
```

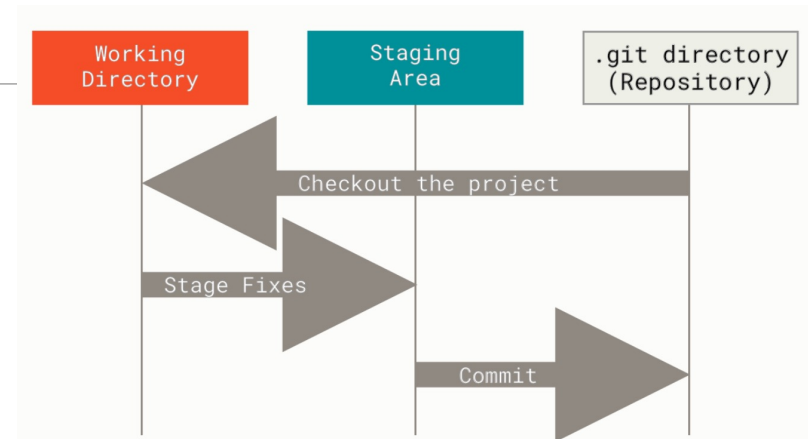
💡 **ProTip!** Use the URL for this page when adding GitHub as a remote.

GitHub Usages

- Sharing code (GitHub)
 - Variants: Sharing code and taking actions – e.g., building code-bases periodically
- Creating and hosting web pages (GitHub pages)

GitHub – Managing Code

- Process for making change
 - Make local changes
 - **Add** – add changes to (local) repo
 - **Commit** – tell git that you intend to make changes final
 - **Push** – tell git to make changes to remote server



Source: **Pro Git**, Scott Chacon and Ben Straub (2014),
<https://git-scm.com/book/en/v2>

Demo – Git, GitHub

- Creating an organization
- Creating a repo
 - Adding license
 - Adding .gitignore
 - Cloning a repo
- Creating file
 - Committing changes
 - Pulling a file

GitHub Pages

GitHub Pages

- Using GitHub to host website
 - If you have an account on GitHub, you can create the associate website
 - Uses Git and well-defined file layout to fetch and manage website pages
- Uses HTML and css
- For more, see external presentations (published with permission)
 - GitHub pages – TyB presentation - [https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class3-class4-ExternalSlides/Week%2003-TA-TyB\(GitNGitHub\).pdf](https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class3-class4-ExternalSlides/Week%2003-TA-TyB(GitNGitHub).pdf)
 - GitHub pages / HTML – RobertB presentation - [https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class3-class4-ExternalSlides/Week%2003-RobertB\(GitHubPagesHTML\).pdf](https://github.com/biplav-s/course-cmw-f26/blob/main/resources/class3-class4-ExternalSlides/Week%2003-RobertB(GitHubPagesHTML).pdf)

Document using Markdown

- Background
 - A simple markup language for the web
- Resources
 - Basics: <https://www.markdownguide.org/basic-syntax/>
 - Cheat sheet: <https://www.markdownguide.org/cheat-sheet/>

Concluding Section

Conclusion

We talked about

- Sharing Resources – Importance and What to Watch Out For
- GitHub
- GitHub pages
- Touched upon HTML and Markdown

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics)	Github account, repo
4	VS Code, HTML basics	None
5	More HTML, CSS	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	Building a resume, LinkedIn	Draft resume in PDF
10	Review resumes	Review resumes, update Github
11	Convert resume to HTML	Begin conversion, update Github
12	Github code repos	Add repos to portfolio, add SDLC outcomes
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	All extensions end

Ask Me Anything
